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Forecasting Seasonal Commodity Prices with Limited Data: Evaluating ETS vs. LSTM Deep Learning Models

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Abstract. *Rice price volatility in West Java, a major national rice producer, critically impacts Indonesia's food security and inflation rates. This study aims to determine the most accurate forecasting method for medium-quality rice prices by comparing a classical statistical approach, Error-Trend-Seasonal (ETS), against a deep learning-based Long Short-Term Memory (LSTM) model. Monthly price data at the milling level from January 2013 to December 2024 were analyzed. The ETS model structure was selected based on the Akaike Information Criterion (AIC), while LSTM hyperparameters were optimized using the Optuna framework to ensure objective comparison. Forecasting accuracy was evaluated using Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) on out-of-sample data. The empirical results demonstrate that the ETS(M,Ad,A) model significantly outperformed the LSTM architecture. The ETS model achieved high accuracy with a MAPE of 3.26%, whereas the LSTM model yielded a MAPE of 5.36%. The superior performance of ETS is attributed to its effectiveness in capturing strong seasonal patterns and damped trends within a limited sample size (144 observations), whereas LSTM struggled with generalization despite hyperparameter tuning. Consequently, the study recommends the ETS approach as a reliable early warning system for policymakers to anticipate price fluctuations and plan market interventions 3 to 6 months in advance.*

Keywords: *Agricultural Commodity Forecasting; Seasonal Volatility; Eror Trend Seasonal; Long Short-Term Memory; Small-Sample Time Series.*

1. Introduction

Maintaining food security and national economic stability is one of the key priorities in Indonesia. As a staple food for the population, fluctuations in rice prices directly affect purchasing power, inflation rates, and the welfare of farmers [1]. The stability of rice prices is also a critical factor in ensuring national food security, particularly for low-income groups. Therefore, monitoring and forecasting rice prices have become a key focus for the government in ensuring the availability and sustainable distribution of food.

West Java Province is one of the major rice granaries in Indonesia, contributing significantly to the country's rice production. According to data, West Java has consistently ranked among the top five rice producers in Indonesia, with a substantial contribution to the national supply [2]. However, the price of medium-quality rice at the milling level has shown fluctuations during the 2013-2024 period, caused by various factors such as planting and harvesting seasons, production costs, the price of Dry Milled Grain (GKG), and Government Purchase Price (HPP) policies [3]. Medium-quality rice was chosen for this study as it is the most

consumed and traded category in the domestic market [4]. Therefore, price changes in this category are believed to impact food inflation and household economic stability.

Price fluctuations at the rice milling level have a significant impact on the food supply chain as they represent the starting point for price formation before reaching the wholesale and retail markets. Increases in milling prices generally push up consumer prices, potentially reducing public purchasing power. Conversely, when milling prices drop significantly, the profit margins of farmers and milling businesses also decrease, which can reduce the motivation to produce [5]. Therefore, an accurate understanding of the price patterns of medium-quality rice at the milling level is crucial for formulating fair, sustainable pricing policies that maintain a balance between producers and consumers.

This research utilizes two forecasting approaches to generate more accurate and adaptive price predictions in accordance with the characteristics of time series data, Error-Trend-Seasonal (ETS) as a classical statistical method and Long Short-Term Memory (LSTM) as a deep learning-based method. The ETS model is chosen for its ability to explicitly capture trend and seasonal patterns, while LSTM excels in recognizing nonlinear patterns and long-term relationships between data periods [6] [7]. The comparison between these two models aims to determine the most effective approach for forecasting medium-quality rice prices at the milling level in West Java, while also providing a scientific basis for the formulation of price stabilization and regional food security policies.

2. Method

This research conducts a rice price process forecasting in West Java through several systematic stages involving selection and application of two models, namely ETS and LSTM. Each stage is designed to ensure that the chosen model can predict rice prices with optimal accuracy by considering trends, seasonality, and long term relationships within the data. The research methodology used in this study consists of several steps, which are presented in figure 1.

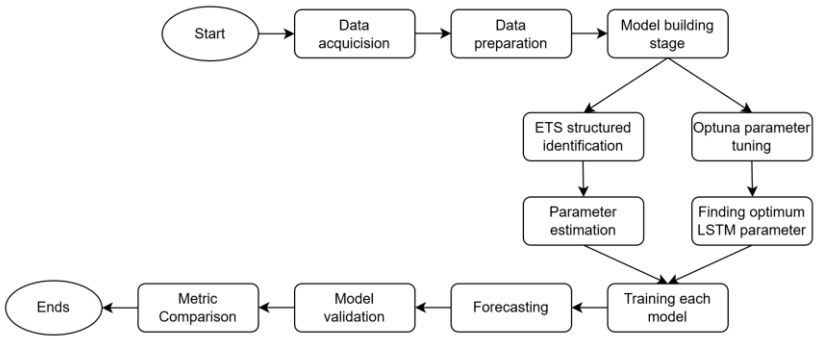


Figure 1. Research methodology flowchart

2.1 Data Acquisition

This study utilizes data on medium-quality rice prices at the milling level in West Java Province, covering the period from January 2013 to December 2024. The data were obtained from the official publications of Statistics Indonesia (BPS) West Java Province through the website

jabar.bps.go.id. The dataset contains monthly average rice prices (in IDR/kg), collected through routine surveys of rice mills across various districts and municipalities. This information reflects price dynamics at the producer level before the rice enters wholesale and retail markets, thereby representing the early stage of rice price formation in the region.

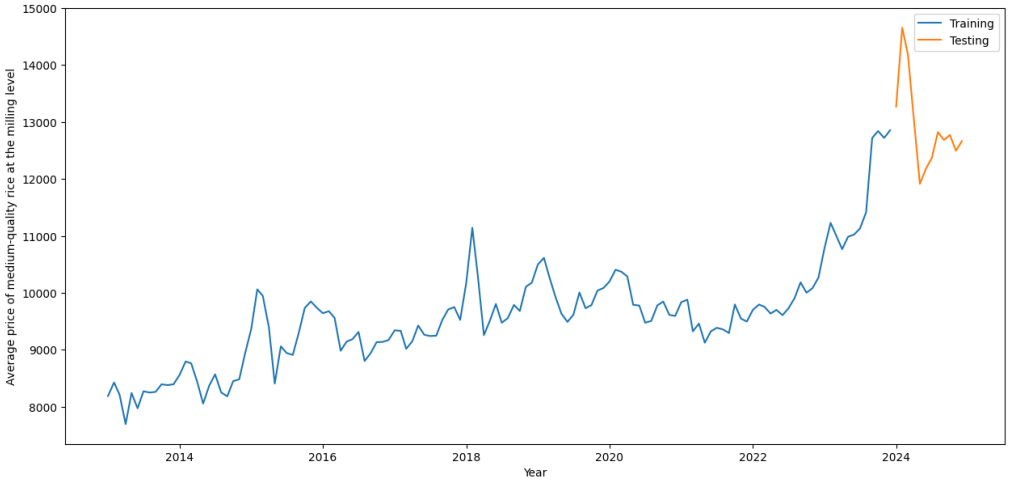


Figure 2. Time series plot of monthly rice prices from 2013 to 2024

The dataset consists of 144 observations, which were then divided into two parts using 11:1 ratio, resulting in 132 training data points and 12 testing data points. The training data were used to construct the forecasting model based on historical price patterns, while the testing data were employed to assess the model's performance on new, unseen data. This split aims to ensure that the resulting model not only fits past patterns but also demonstrates good generalization capability in predicting future rice prices [8].

2.2 Research Methods

2.2.1 ETS Model (Exponential Smoothing State Space Model)

The ETS (Error–Trend–Seasonal) model is a general framework for exponential smoothing that handles data with level, trend, and seasonal patterns simultaneously. This model includes and extends the SES, DES, and TES approaches as special cases. In the ETS framework, the error component (E) can be additive (A) or multiplicative (M); the trend component (T) can be none (N), additive (A), or additive damped (Ad); and the seasonal component (S) can be none (N), additive (A), or multiplicative (M) [9].

The stationary model in the exponential smoothing method is represented by the single exponential smoothing (SES) approach as the most basic form of smoothing.

$$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1}) \quad (1)$$

Explanation:

F_t : new forecast

F_{t-1} : previous forecast

α : smoothing coefficient $(0 \leq \alpha \leq 1)$

A_{t-1} : actual value for the previous period

The trend model in the exponential smoothing method is represented through the double exponential smoothing (DES) approach, which was developed based on Brown's formulation and

Holt's trend modeling.

$$A_t = \alpha Y_t + (1-\alpha)(A_{t-1} + T_{t-1}) \quad (2)$$

$$T_t = \beta(A_t - A_{t-1}) + (1-\beta)T_{t-1} \quad (3)$$

$$\hat{Y}_{t+p} = A_t + T_t p \quad (4)$$

Explanation:

Y_t : actual data for period t
A_t : exponential smoothing value
α : non-trend smoothing constant parameter
β :smoothing parameter for trend estimation
T_t : trend model estimation
$\hat{Y}$: forecast results for future periods
p : number of periods forecast

The seasonal model in the exponential smoothing method, known as triple exponential smoothing (TES) or the Winters model, consists of two main forms, namely the additive approach and the multiplicative approach. The additive seasonal form can be expressed as follows:

$$\hat{y}_{T+p}(T) = \ell_T + pb_T + sn_{T+p-L} \qquad (p = 1,2,3,\dots) \quad (5)$$

$$\ell_T = \alpha(y_T - sn_{T-L}) + (1-\alpha)(\ell_{T-1} + b_{T-1)} \quad (6)$$

$$b_T = \gamma(\ell_T - \ell_{T-1}) + (1-\gamma)b_{T-1} \quad (7)$$

$$sn_T = \delta(y_T - \ell_T) + (1-\delta)sn_{T-L} \quad (8)$$

Explanations:

ℓ_T : exponential smoothing estimate
b_T : trend estimate
sn_T : seasonal estimate
α : smoothing coefficient
γ : trend coefficient
δ : seasonal coefficient
Values α, γ and δ between 0 and 1
L : seasonal length

The selection of the best ETS model was carried out by comparing several model structures using the Akaike Information Criterion (AIC). The model with the smallest AIC value was considered the most optimal because it provided the best balance between model fit and parameter complexity [10]. After the model is selected, a residual diagnostic check is performed using the Ljung–Box test to ensure that the residuals are white noise and do not show significant autocorrelation [11]. The stages of ETS model analysis are as follows:

1. Exploration of Time Series Patterns
   The first stage was conducted by analyzing rice price time series patterns to identify long-term trends, seasonal patterns, and random fluctuations. This visualization helped determine whether ETS components were relevant to the data being analyzed.
2. Division of Data into Training Data and Test Data
   The data is divided based on time periods, with training data set from January 2013 to

December 2023, and test data from January 2024 to December 2024. The year-based division is used to maintain the chronological order of the time series and ensure that the model is tested on a complete seasonal cycle. This approach also prevents data leakage, as seasonal patterns within a year are not interrupted by data division, resulting in more realistic validation results that reflect actual forecasting scenarios.

3. Stationarity Test
   Next, a test is conducted to determine whether the data is stationary using the Augmented Dickey–Fuller (ADF) test. This test is conducted to determine whether the train data has a trend component or not. If the data is not stationary in mean, then the data has a trend, vice versa.
4. Determination of ETS Model Candidates
   ETS model candidates are obtained by looking at the presence or absence of a trend component based on the results of the stationarity test. If there is no trend or stationarity is satisfied, then the Trend (T) component can contain none (N), additive (A), or additive damped (Ad). Meanwhile, if there is a trend, the Trend (T) component cannot contain none (N).
5. Ljung-Box Test and Elimination of ETS Model Candidates
   The Ljung-Box test on ETS model candidates is performed to ensure that the model residuals resemble white noise. If there are models that do not meet this assumption, they will be eliminated from the model candidates that will be compared next.
6. Determining the ETS Model Structure
   The next step is to determine the appropriate ETS model structure. Several combinations of Error (E), Trend (T), and Seasonal (S) components, such as ETS(A,A,A) or ETS (M,A,M), are evaluated. Model selection is performed automatically using the ets() function, which compares various model structures based on the Akaike Information Criterion (AIC), MAPE, and RMSE values.
7. Parameter Estimation
   The selected model is then estimated using the Maximum Likelihood Estimation (MLE) method. The smoothing parameters α, β, and are updated to minimize errors in historical data so that the model can capture patterns more accurately [12].
8. Forecasting Using the Selected Model
   The best ETS model is used to generate forecasts and prediction intervals. In the state-space framework, prediction intervals are obtained automatically, providing an overview of the uncertainty of the forecast results.
9. Out-of-Sample Validation
   The final stage is to evaluate the model's performance by comparing the forecast results with the test data using indicators such as MAPE, RMSE, and MAE. This validation ensures the ETS model's ability to predict new data and assess the accuracy of the forecast.

2.2.2 LSTM Model (Long Short-Term Memory) with Optuna Optimization

LSTM is a specialized architecture of Recurrent Neural Network (RNN) developed to overcome the limitations of traditional RNN, particularly the vanishing gradient problem and the inability to learn long-term dependencies. Unlike stochastic statistical models such as ARIMA, which assume linear relationships, LSTM has superior capabilities in modeling complex non-linear patterns [13].

In the context of volatile monthly rice price data, LSTM is highly relevant due to its ability to capture separate trend and seasonal components separated by long time-lags through the cell state mechanism. This internal memory capability allows the model to retain important information from past historical sequences and ignore noise.

1. Data Preprocessing

   Preprocessing steps are performed systematically to ensure that raw data is ready for consumption by the LSTM algorithm. Data is sorted by time index and normalized to the range [0,1] to ensure numerical stability and accelerate the convergence of the gradient optimization algorithm using the equation:

$$x' = \frac{x - min(x)}{max(x) - min(x)} \tag{9}$$

   Time series data is converted into supervised learning format using the sliding window method. Historical data for n months (lookback period) is used as input (X_t) to predict the value for the following month (Y_{t+1})

2. LSTM Network Architecture

   LSTM has a gating system mechanism consisting of an input gate, forget gate, and output gate that control the flow of information in memory cells [14]. This mechanism allows the network to selectively forget irrelevant information and store important information for a long period of time.

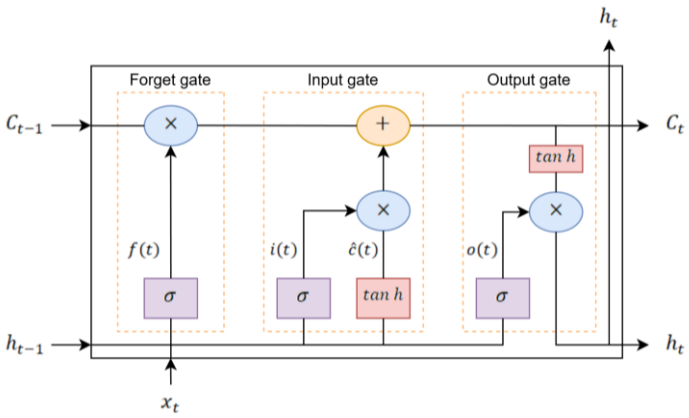


Figure 3. Illustration of single-cell long short-term memory architecture

The forget gate determines which information from the previous cell state should be discarded or retained. This gate applies a sigmoid function to the previous hidden state and the current input, outputting a value between 0 and 1 for each number in the cell state C_{t-1}.

$$f_t = \sigma\left(W_f[h_{t-1}, x_t] + b_f\right) \tag{10}$$

The input gate controls the extent to which new information flows into the cell state. It employs a sigmoid layer to decide which values to update and a tanh layer to create a vector of new candidate values ($\hat{C}_t$), which are then combined to update the cell state.

$$i_t = \sigma\left(W_i[h_{t-1}, x_t] + b_i\right) \tag{11}$$

$$\hat{C}_t = tanh\left(W_C[h_{t-1}, x_t] + b_C\right) \tag{12}$$

The output gate regulates what part of the cell state makes it to the final output based on the

filtered cell state. The current cell state is passed through a tanh function to push values between -1 and 1, and then multiplied by the output of the sigmoid gate to produce the final hidden state.

$$o_t = \sigma\left(W_o[h_{t-1}, x_t] + b_o\right) \quad (13)$$

$$h_t = o_t \times tanh(C_t) \quad (14)$$

The cell state is subsequently updated using the results from the forget and input gates

$$C_t = f_t \times C_{t-1} + i_t \times \hat{C}_t \quad (15)$$

The variables depicted in the architecture diagram are defined as follows:

x_t : The input vector at the current time step *t*

h_{t-1} : The hidden state vector from the previous time step, carrying short-term memory information

C_{t-1} : The cell state from the previous time step, representing the long-term memory of the network.

C_t : The updated cell state at the current time step, resulting from the combination of the forget gate and input gate operations.

h_t : The output hidden state at the current time step, which will be passed to the next LSTM cell or the final output layer

3. Optuna Hyperparameter Optimization

Optimization was performed using the Optuna framework with the Tree-structured Parzen Estimator (TPESampler) search strategy to identify the optimal hyperparameter combination that minimizes forecasting error. TPE works by modeling the probabilities $p(x|y)$ and $p(y)$ to select the next hyperparameter configuration that is most likely to improve model performance based on previous trial history.

The objective function used is to minimize the Mean Absolute Percentage Error (MAPE) value calculated on the test data, and the model is compiled using the Mean Squared Error (MSE) loss function and Adam Optimizer. The final LSTM model is retrained using the best hyperparameter combination found by Optuna.

In this study, the hyperparameter search space optimized by Optuna includes:

a. Neurons: Determines the complexity of the model's memory capacity, ranging from 32 to 64 units.
b. Learning Rate: Controls the magnitude of weight updates during training ranging from 0.0001 to 0.01.
c. Time Steps: Determines the length of the input sequence, ranging from 3 to 12 steps.
d. Batch Size: Determines the number of samples processed before updating the model's internal parameters between 16 and 32.

2.3 Indicator Assesment

There are three main indicators for evaluating the performance of ETS and LSTM models in forecasting rice prices in West Java, namely MAPE, RMSE, and MAE. These indicators were chosen because each provides a different perspective in assessing the accuracy and generalization ability of the model. MAPE measures the relative error between the predicted value and the actual value as a percentage. It is important to recognize that the acceptable threshold for MAPE may vary depending on the specific context of the data. Generally, a MAPE value below 10% is considered satisfactory for forecasting models, while values below 5% are regarded as excellent.

However, these thresholds can be adjusted based on factors such as the volatility and the nature of the data being analyzed. The formula for MAPE is:

$$MAPE = \frac{100\%}{n}\sum_{t=1}^{n}\left|\frac{y_t - \widehat{y_t}}{y_t}\right| \quad (16)$$

RMSE measures the magnitude of the average squared error between the prediction and the actual values. A lower RMSE indicates that the model is better at aligning its predictions with the observed data. It is crucial to note that RMSE is highly sensitive to the scale of the data. Larger values in the dataset will result in higher RMSE values, making the measure more responsive to large variations in the data. The RMSE formula is:

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}|y_t - \widehat{y_t}|^2} \quad (17)$$

MAE measures the average absolute error between the predicted value and the actual values. The lowest MAE indicates a better model for predicting overall rice prices. Similar to RMSE, the MAE values are influenced by the magnitude of the data being predicted. For instance, accurate predictions of large values will result in a lower MAE. However, MAE is less sensitive to outliers compared to RMSE, making it a more robust measure when extreme values are present in the dataset. MAE formula is:

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \widehat{y_t}| \quad (18)$$

3. Results and Discussion

3.1 Data Exploration

Time series analysis is performed to understand historical patterns before modeling. This step is important because time series data characteristics, such as trends and seasonal patterns, can affect the performance of the forecasting method used. Therefore, a decomposition process is applied to separate rice price variability into its main components so that patterns of change can be observed more clearly. Based on the results of the time series decomposition in figure 4, monthly rice price data is broken down into three main components: trend, seasonal, and residual.

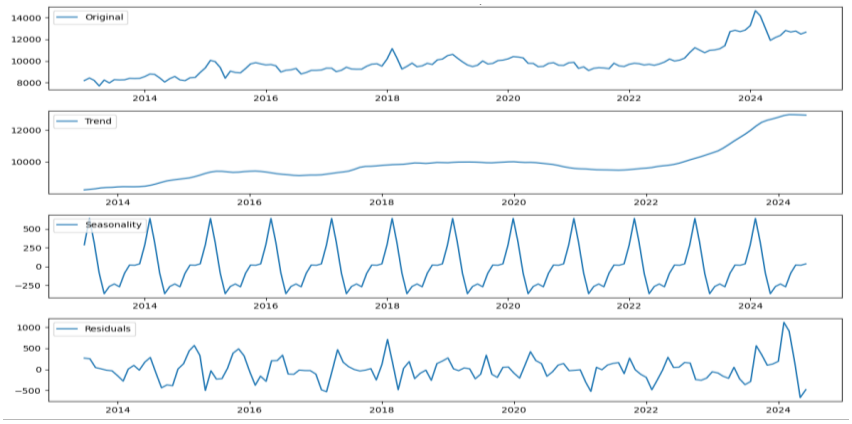


Figure 4. Decomposition of monthly rice price data

The time series decomposition indicates that the dynamics of rice prices in West Java are dominated by a long-term positive trend that experienced a significant structural surge during the

2023–2024 period. The seasonal pattern is stable, displaying a recurring 12-month cycle that reflects planting and harvest periods, while the residual component shows increasing volatility (heteroskedasticity) in recent years. This non-stationary trend and heightened residual fluctuations confirm the complexity of the data, which requires stationarity handling through differencing techniques, as well as the use of non-linear models such as LSTM to achieve optimal forecasting accuracy.

3.2 ETS Results

ETS model analysis was conducted to obtain a modeling structure capable of accurately describing trend and seasonal patterns in rice price data. The steps taken included model candidate selection, parameter estimation, residual examination, and prediction accuracy validation using test data. This approach ensures that the model is not only suitable for historical data, but also capable of generalization in the prediction period. The time series pattern exploration stage has been discussed in the data exploration section, so the ETS results in this section focus on the modeling stages, from stationarity checks to validation.

The data was divided into training data and test data in order to separate the model building and evaluation processes. The ADF test performed on the training data produced a p-value of 0.9231, indicating that the time series was non-stationary and had a trend component. This condition forms the basis for selecting suitable model candidates, namely ETS(A,A,A), ETS(M,A,A), ETS(M,A,M), ETS(A,Ad,A), ETS(M,Ad,A), and ETS(M,Ad,M). All model candidates were then tested using the Ljung-Box test to ensure that the residuals met the white noise assumption before proceeding to the next selection stage.

Table 1. Ljung-box test results

Model	P-value	Decision
ETS(A,A,A)	0.509	Accept H0
ETS(M,A,A)	0.555	Accept H0
ETS(M,A,M)	1.878e-05	Reject H0
ETS(A,Ad,A)	0.4615	Accept H0
ETS(M,Ad,A)	0.4073	Accept H0

The Ljung-Box test results show that the ETS(M,A,M) model does not meet the residual assumption and is therefore eliminated from the model candidates. The remaining models are then compared using AIC values and initial performance indicators such as MAPE and RMSE. This evaluation is conducted to assess the balance between model fitness and the ability to predict new data. All comparison results are shown in the following table as a basis for further model selection.

Table 2. Comparison of ETS models

Model	AIC	MAPE	RMSE
ETS(A,A,A)	2167.149	2.217	281.099
ETS(M,A,A)	2169.191	2.275	287.197
ETS(A,Ad,A)	2165.992	2.167	277.758
ETS(M,Ad,A)	2162.152	2.161	278.348
ETS(M,Ad,M)	2155.325	2.109	271.016

Referring to the comparison results in Table 2, the ETS(M,Ad,M) model has the lowest AIC value, but its performance declines in the test data, indicating overfitting. The ETS(M,Ad,A) model provides the best balance between AIC, MAPE, and RMSE values, both in the training data and test data. This condition shows that this model is not only suitable for historical data, but also has more stable predictive capabilities.

Beyond empirical performance, the selection of ETS(M,Ad,A) is also supported by the underlying characteristics of staple food prices. Damped trend models are often more realistic for economic time series, as long-term growth cannot increase indefinitely [9]. Such behavior is commonly observed in food price dynamics, where upward trends tend to moderate due to government price-stabilization policies, namely the SPHP program administered by BAPANAS, which aims to prevent excessive price increases through supply-side interventions. Therefore, the ETS(M,Ad,A) model is not only statistically optimal but also substantively aligned with the economic mechanism governing rice price dynamics.

The selection of the model was then followed by parameter estimation analysis to understand the level, trend, and seasonal components. Parameter estimation in the ETS(M,Ad,A) model resulted in an α value of 0.7858, very small β and γ values of 0.0001, and a φ value of 0.98. The high α value indicates that the model is highly responsive to recent changes in the data, while the small β and γ values reflect a slowly developing trend and a stable seasonal pattern. The φ value, which is close to one, indicates a dampened trend that still influences price movements over time.

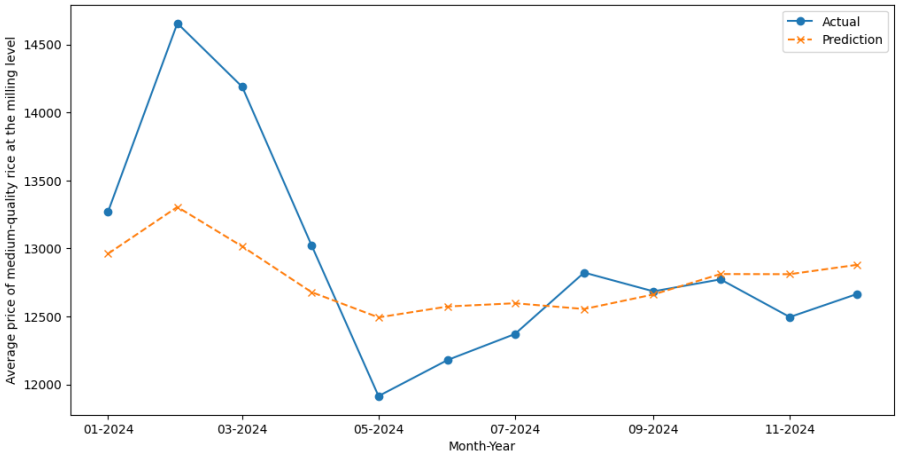


Figure 5. Comparison plot between ETS forecast and actual data

The ETS(M,Ad,A) model was then used to forecast rice prices for the period January to December 2024. The visualization results show that the model successfully followed seasonal patterns consistently and the predictions remained within a stable confidence interval range. Although there was a slight delay in sharp price increases, the model was still able to accurately capture the main patterns. This shows that the seasonal mechanism in the model works well in accommodating recurring price variations each year.

Model performance was evaluated using three metrics, namely MAPE, RMSE, and MAE. The evaluation results show that the ETS(M,Ad,A) model has a MAPE value of 3.26%, which indicates a relatively low prediction error rate. In addition, the RMSE value of 589.92 and MAE

of 436.41 indicate that the deviation between the actual value and the prediction result is still within acceptable limits for commodity price data, which tends to fluctuate. These findings reinforce that the ETS(M,Ad,A) model provides stable prediction performance and has adequate accuracy in projecting rice price movements.

3.3 LSTM Results

Referring to the methodological framework described earlier, the LSTM model training process does not use static parameters, but rather searches the parameter space (hyperparameter search space) using the TPE algorithm in Optuna. This approach is applied to obtain the best parameter configuration that balances model complexity with generalization capabilities on new data. After undergoing a series of trials, the model reached an optimal loss convergence point with a specific combination of parameters. This final configuration was used as the basis for model development and training prior to performance evaluation.

The resulting model has a look-back period of 24 months, which indicates that in order to accurately predict next month's rice prices, LSTM requires historical context spanning two full calendar cycles. This length of time window allows the internal memory mechanism to capture annual seasonal patterns, such as harvest cycles and lean periods, as well as medium-term trends that cannot be captured by shorter time windows such as three or six months. In addition, the optimal configuration includes 56 neurons in the hidden layer, which is considered to be at an equilibrium point because it is complex enough to learn nonlinear patterns without increasing the risk of overfitting. The model also uses a learning rate of 0.0004 through the Adam optimizer, where this value allows weight updates to take place gradually and stably so that the model can reach the global minimum on the Mean Squared Error (MSE) loss function.

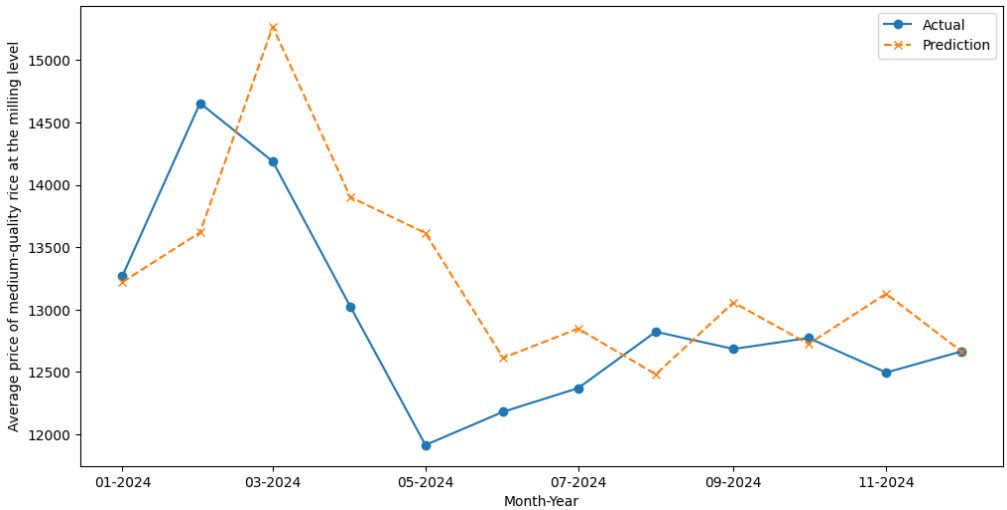


Figure 6. Comparison plot between LSTM forecast and actual data

A quantitative evaluation was conducted using test data from January to December 2024 and produced a MAPE value of 4.54%. This value indicates that the model has good predictive capabilities with a relatively small average deviation from actual prices, so it can be said that the model is able to capture the dynamics of rice prices in West Java. Visually (Figure X), the forecasting results show that the model not only follows general trend patterns but is also able to

respond to inflection points when extreme price fluctuations occur. The model's ability to map nonlinear patterns and seasonal volatility indicates that the LSTM architecture can accommodate the characteristics of complex and non-linear time series data.

3.4 Model Comparison

The evaluation was conducted using several metrics, namely MAPE, RMSE, and MAE, to obtain a more comprehensive picture of the performance of both models. In addition, the MAPE column in the training data was added as a basis for comparison between performance in the training data and test data, thereby enabling the identification of potential overfitting, especially in the LSTM model, which has a more complex architecture. Table 3 presents a summary of the evaluation results for each model based on these metrics.

Table 3. Comparison of evaluation metrics between the ETS and LSTM models

Model	Evaluation Metrics			
	MAPE Test	MAPE Train	RMSE Test	MAE Test
ETS	3.26%	2.16%	589.92	436.41
LSTM	4.54%	2.30%	762.37	587.40

A comparison of model performance between ETS and LSTM was conducted using four evaluation indicators, namely MAPE on training data, MAPE on test data, RMSE, and MAE. Based on the evaluation results, the ETS model obtained a MAPE value of 2.16% on the training data and 3.26% on the test data, while the LSTM model recorded a MAPE value of 2.30% on the training data and increased to 4.54% on the test data. The difference between MAPE on the training data and test data shows a more stable error pattern in the ETS model compared to LSTM, which experienced a greater increase in prediction errors when applied to the test data.

The larger error difference in the LSTM model indicates the potential for overfitting, especially considering the limited size of the dataset and the complexity of the deep learning architecture. Overfitting occurs when the model learns patterns in the training data too specifically, making it less capable of generalizing to new data. This is also evident from the RMSE value of 762.37 and MAE of 587.40 in LSTM, which are higher than ETS with RMSE values of 589.92 and MAE of 436.41. Thus, the evaluation results show that ETS can provide more stable and accurate prediction performance than LSTM, both in the training and testing phases.

These findings also support the interpretation that monthly rice price data characteristics, which have consistent seasonal patterns and relatively smooth trends, are better modeled using a structural component-based approach such as ETS than a complex non-linear approach such as LSTM. In datasets with limited size and moderate fluctuations such as this study, simple models such as ETS provide better generalization, while LSTM requires a larger volume of data to avoid overfitting on the training data.

4. Conclusion

The classical ETS model proved to be superior to the deep learning LSTM model in forecasting the price of medium-quality rice at the milling level in West Java Province. This is not due to inherent flaws of LSTM, but rather because the model struggles with small datasets,

where its complex architecture tends to overfit the data. In contrast, the parsimonious structure of the ETS model makes it more efficient for small sample sizes, effectively capturing the damped trend and seasonal cycles present in the data. This highlights that, for small datasets, classical statistical methods such as ETS are often more reliable and efficient compared to deep learning models, which perform better on larger, more complex, non-linear datasets.

Specifically, the ETS(M,Ad,A) model produced a lower error rate with a MAPE of 3.26%, compared to LSTM which recorded a MAPE of 4.54%, confirming that for univariate time series data with a limited number of observations (144 months) and strong seasonal patterns, the state space approach is able to capture the damped trend and harvest cycle components more accurately and stably. Conversely, although LSTM has the ability to capture complex non-linear patterns, this model experiences generalization difficulties due to limited historical data, so that high computational complexity does not necessarily result in better prediction accuracy in the context of relatively structured food commodity price volatility.

Based on these findings, further research should not rely solely on historical price data (univariate), but should apply a multivariate approach by including relevant exogenous variables such as rainfall, fertilizer prices, and harvest area to improve the learning capacity of the LSTM model to be more competitive. For policymakers, particularly the West Java Provincial Government and Bulog, it is recommended to adopt the forecasting results from the ETS model as a reliable early warning system in planning grain absorption volumes and market operations at least 3 to 6 months in advance, so that price spikes can be anticipated before they impact regional inflation. Meanwhile, for the community and milling businesses, the trend patterns generated by this model can be used as a strategic reference for stock management and determining the optimal time to buy and sell in order to maintain profit margins and sustainable purchasing power.

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